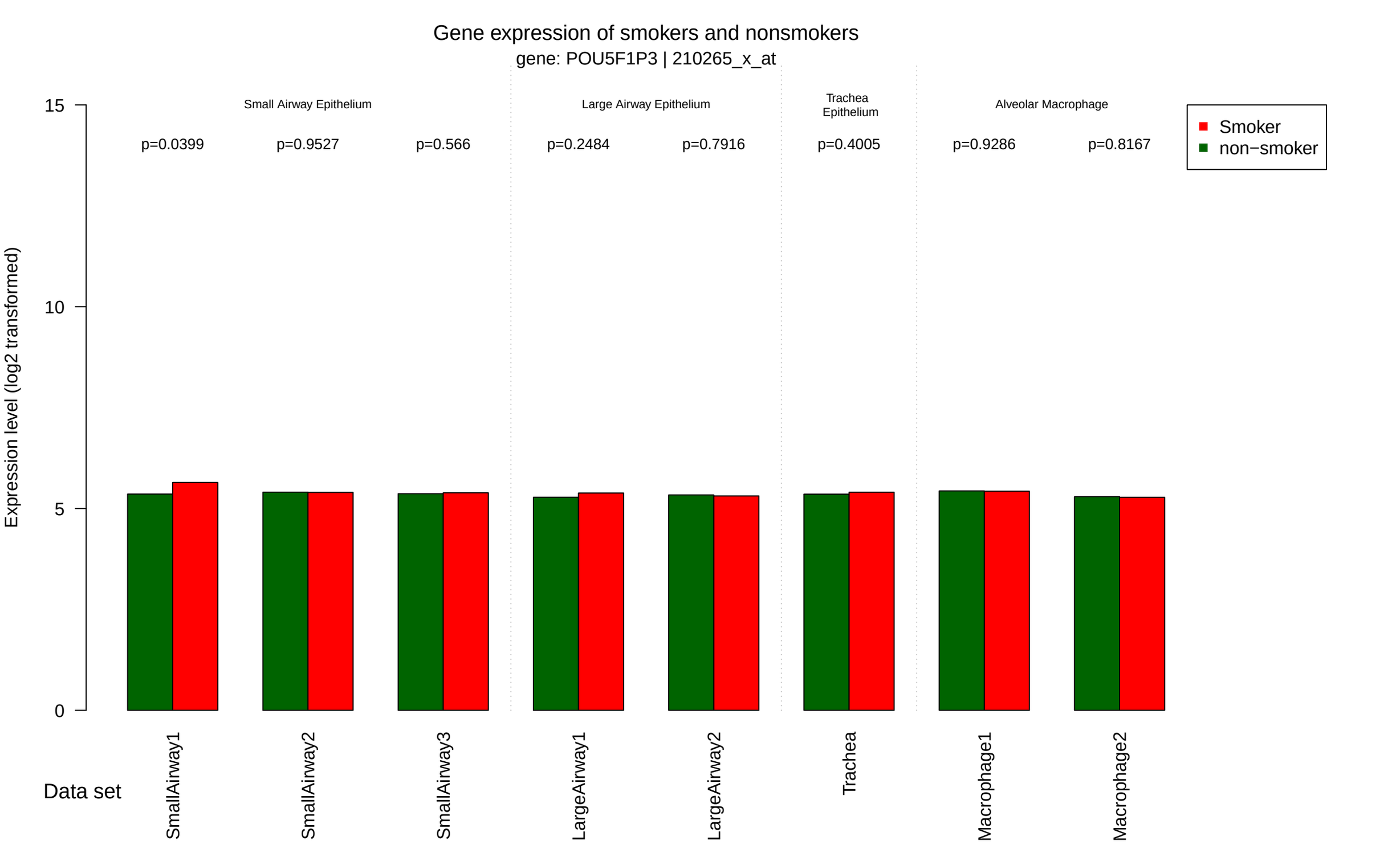
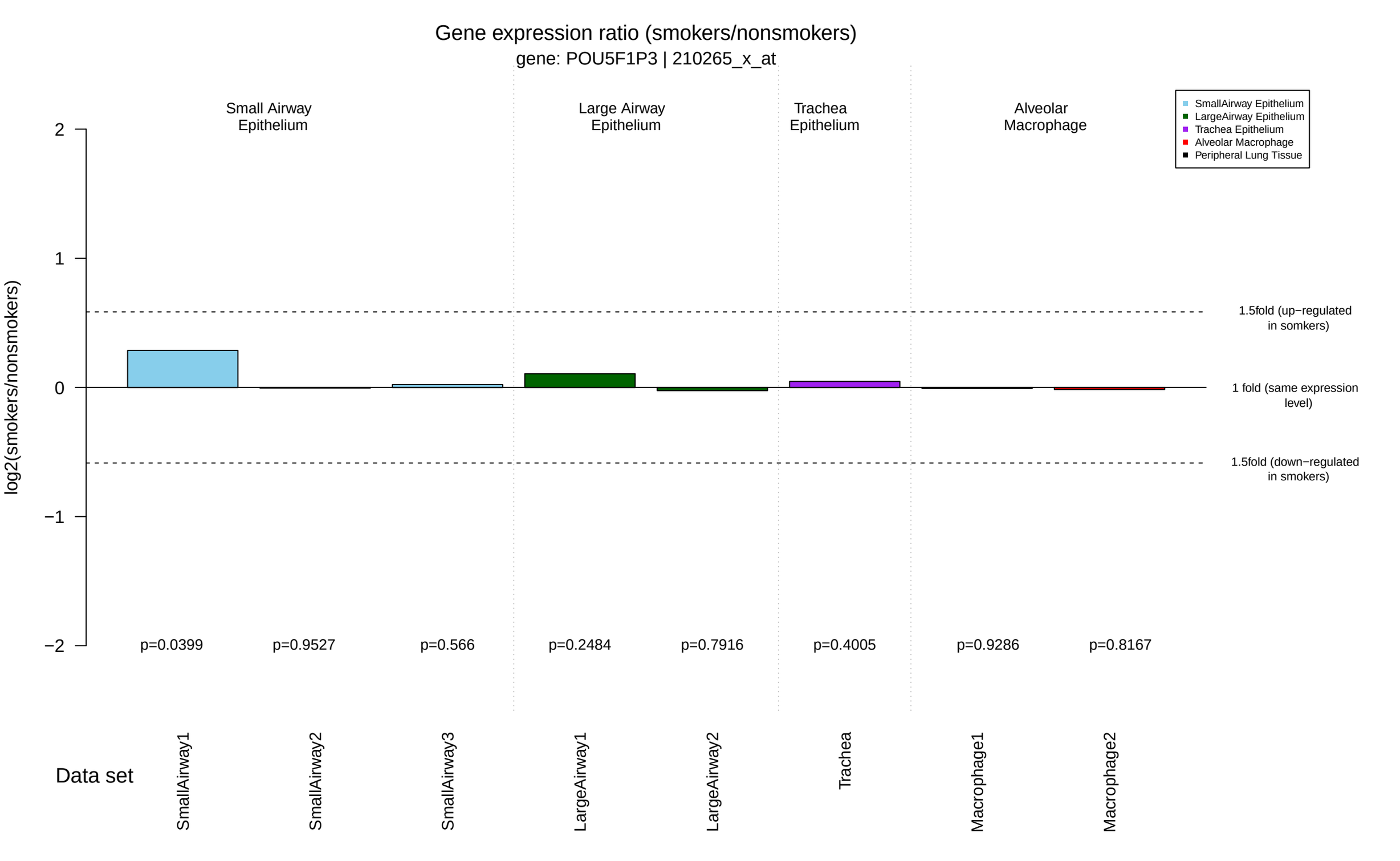
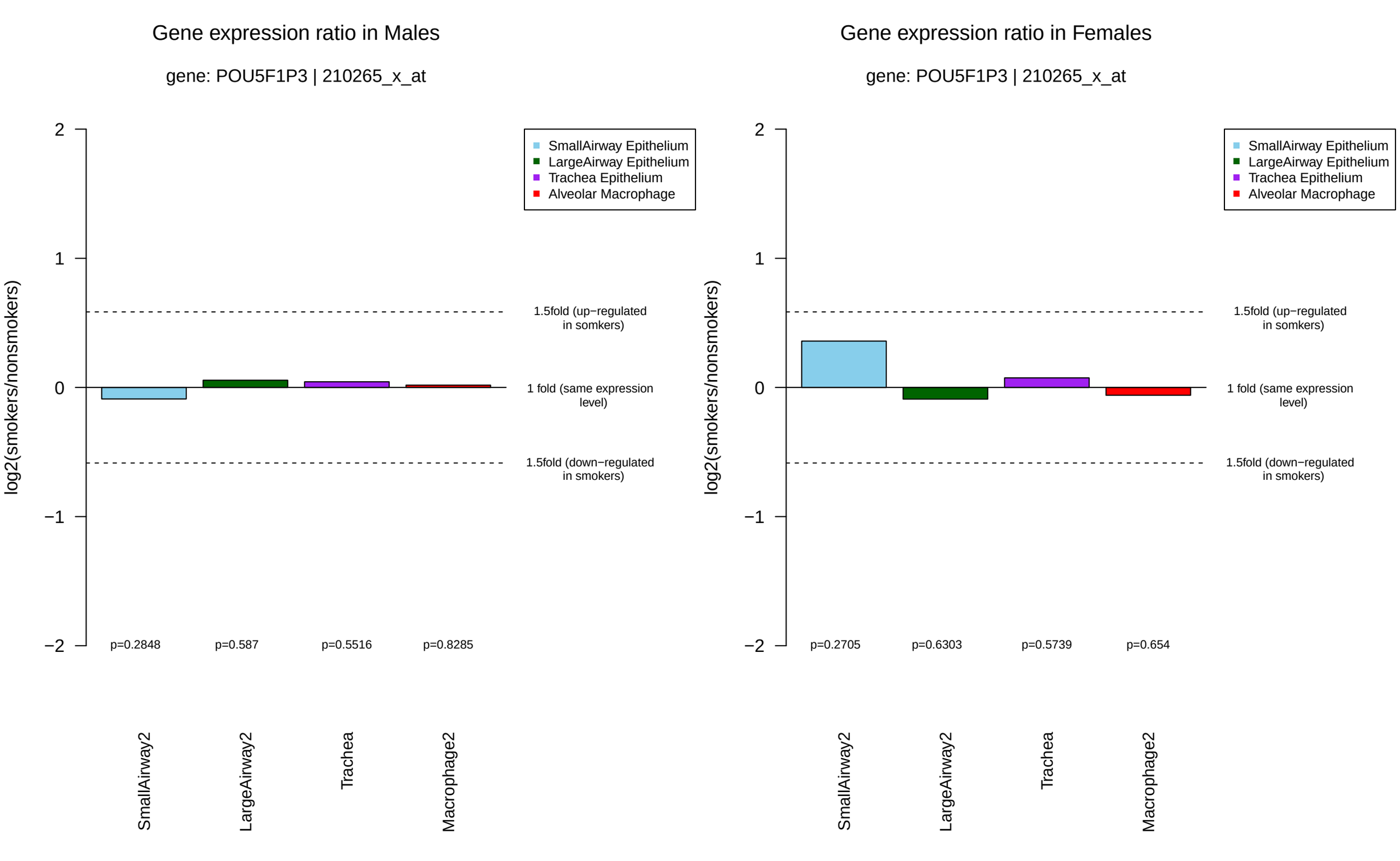
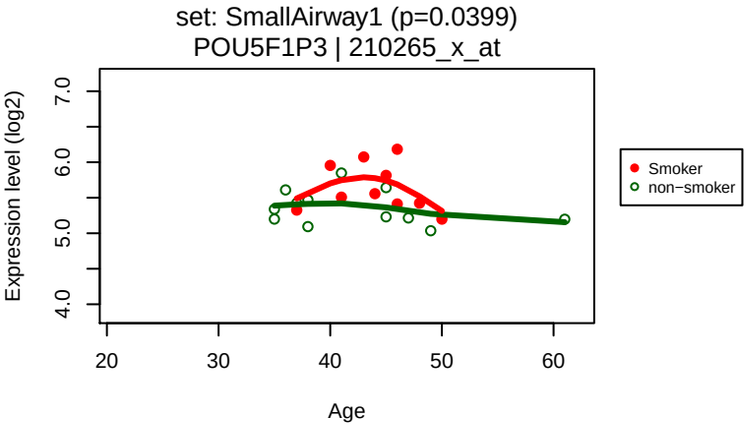
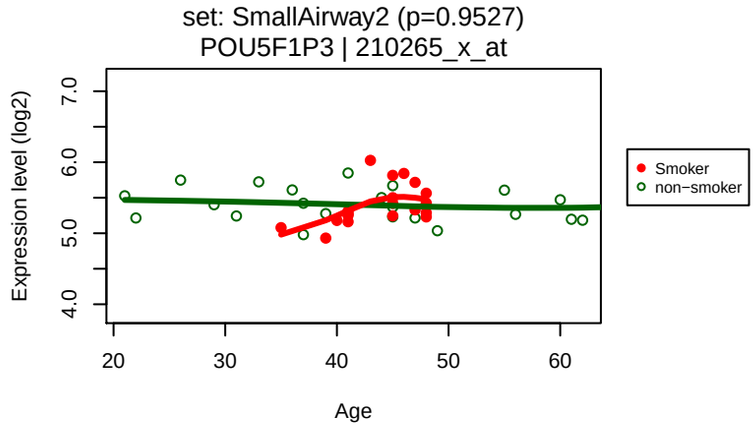
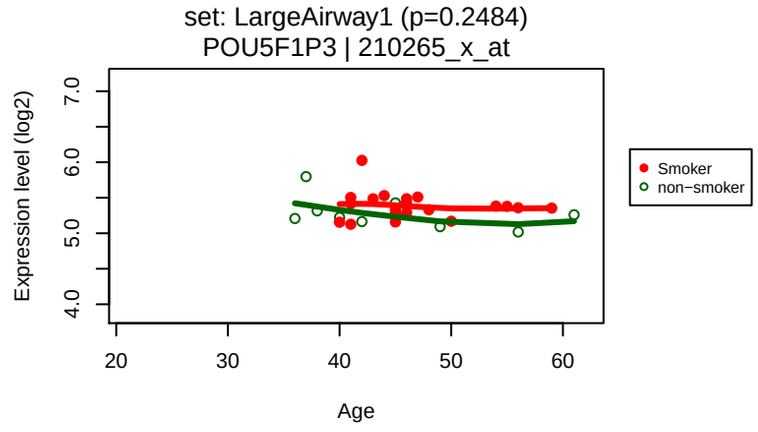

Gene expression of smokers and nonsmokers
gene: POU5F1P3 | 210265_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0399
p=0.9527
p=0.566
p=0.2484
p=0.7916
p=0.4005
p=0.9286
p=0.8167
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

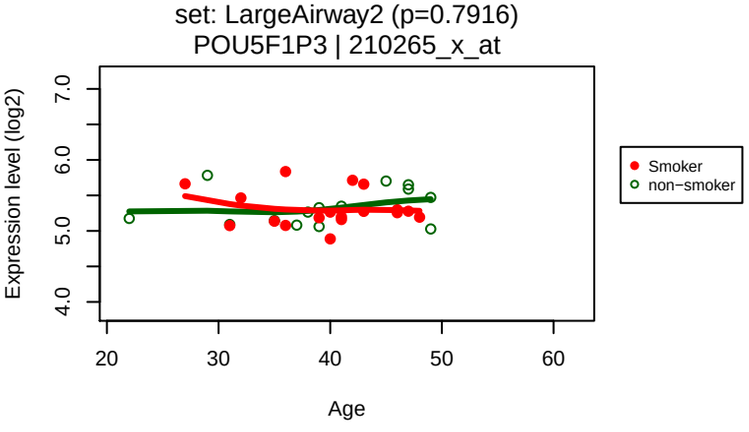

Gene expression ratio (smokers/nonsmokers)
gene: POU5F1P3 | 210265_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0399
p=0.9527
p=0.566
p=0.2484
p=0.7916
p=0.4005
p=0.9286
p=0.8167
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

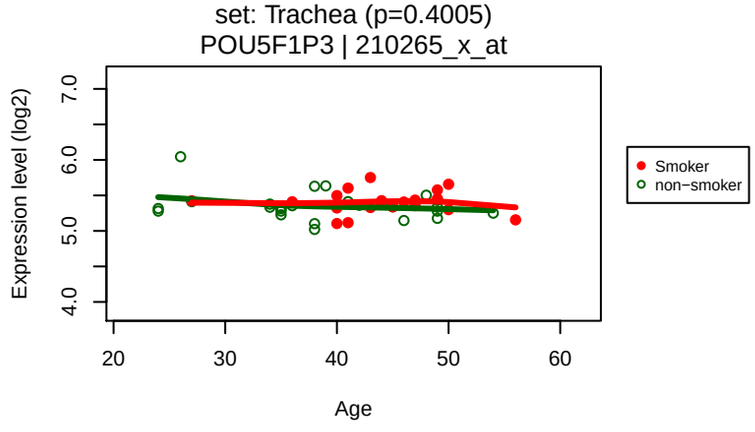

Gene expression ratio in Males
gene: POU5F1P3 | 210265_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2848
p=0.587
p=0.5516
p=0.8285
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: POU5F1P3 | 210265_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2705
p=0.6303
p=0.5739
p=0.654
SmallAirway2
LargeAirway2
Trachea
Macrophage2

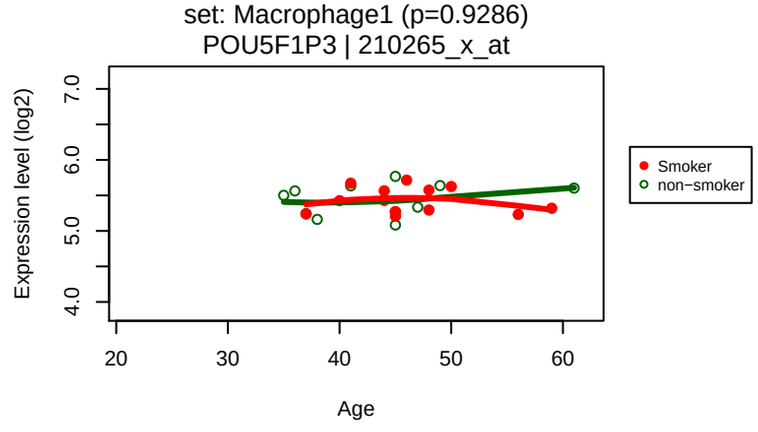

set: SmallAirway1 (p=0.0399)
POU5F1P3 | 210265_x_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker

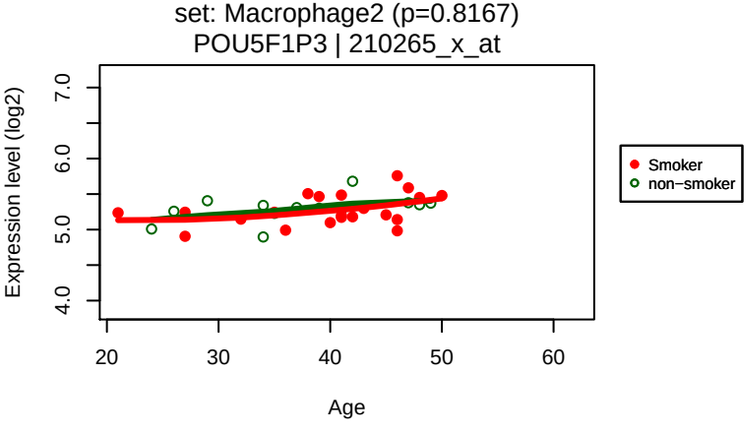

set: SmallAirway2 (p=0.9527)
POU5F1P3 | 210265_x_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker

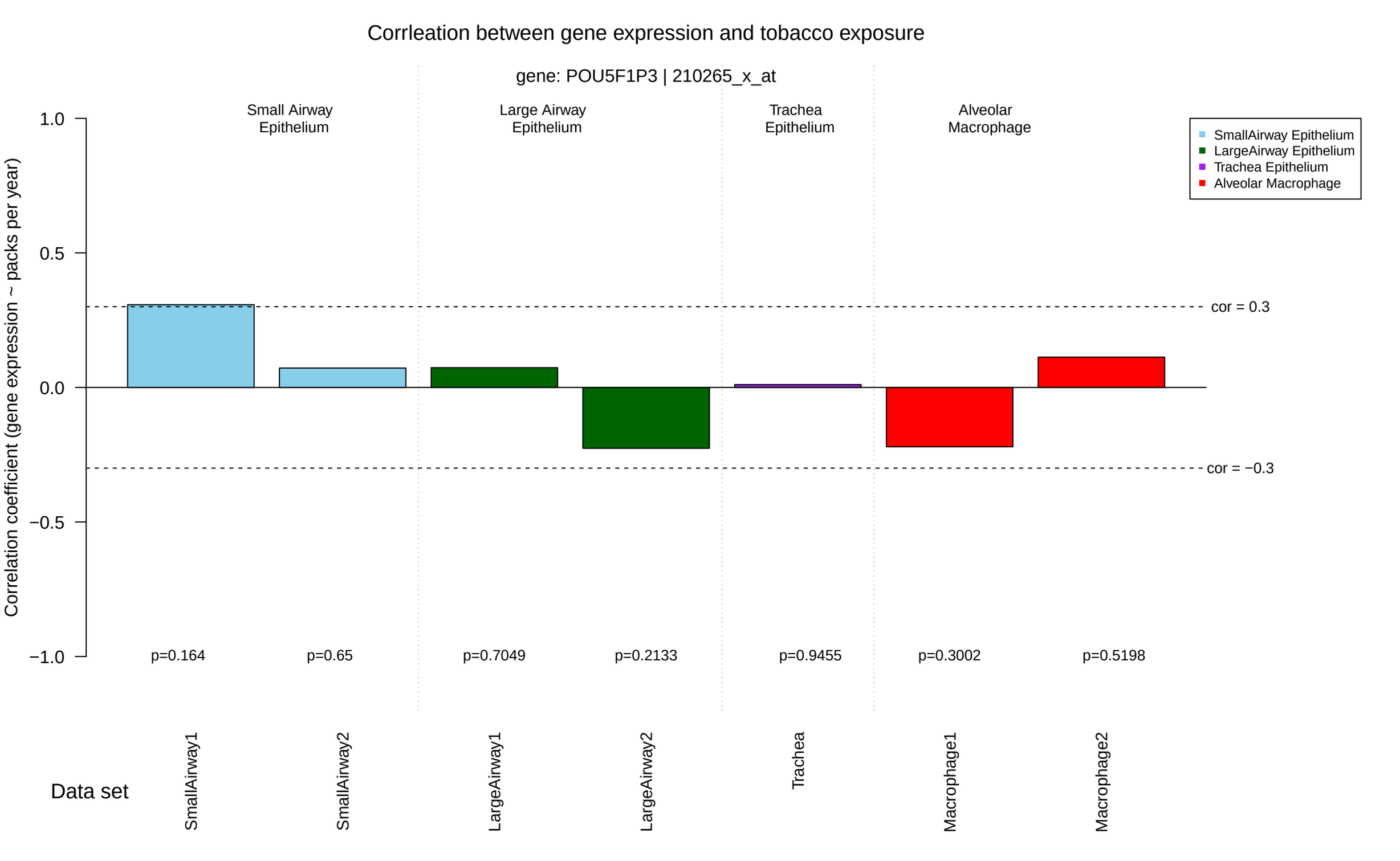

set: LargeAirway1 (p=0.2484)
POU5F1P3 | 210265_x_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7916)
POU5F1P3 | 210265_x_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.4005)
POU5F1P3 | 210265_x_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9286)
POU5F1P3 | 210265_x_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8167)
POU5F1P3 | 210265_x_at
Expression level (log2)
4.0
5.0
6.0
7.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: POU5F1P3 | 210265_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.164
p=0.65
p=0.7049
p=0.2133
p=0.9455
p=0.3002
p=0.5198
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2